Self-Supervised Learning for Mechanical Grey Swan Prediction

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Abstract

Mechanical component failures are *grey swans* - rare but physically plausible events that follow known degradation physics yet are critically underrepresented in training data. Existing supervised prognostic methods require extensive labeled failure trajectories that are expensive and dangerous to collect, limiting their applicability to novel operating conditions. We introduce a self-supervised framework that learns degradation-aware representations from abundant unlabeled vibration data using two complementary pretraining strategies: (1) a Joint Embedding Predictive Architecture (JEPA) that captures waveform texture through masked patch prediction, and (2) temporal contrastive learning that captures spectral centroid shift - the dominant degradation indicator - by exploiting the temporal ordering of run-to-failure episodes. A hybrid architecture fusing JEPA representations with handcrafted spectral features achieves RMSE $= 0.055 \pm 0.004$ on bearing remaining useful life prediction, a 75.5% reduction versus time-only baselines and 21.4% versus the best handcrafted-only model (5 seeds). For cross-dataset transfer (FEMTO→XJTU-SY), temporal contrastive pretraining yields 38% improvement over time-only baselines ($p < 0.001$, 10 seeds). Mechanistic analysis reveals that JEPA and contrastive encoders learn fundamentally different features - waveform periodicity versus spectral dynamics - explaining their complementary strengths.

1 Introduction

A turbine blade cracks after 12,000 flight hours. A bearing seizes in a wind turbine gearbox during peak generation. A pump seal fails in a chemical plant, triggering an emergency shutdown. These are *grey swans* - events that are rare in operational data but follow known degradation physics and are, in principle, predictable (**?**). Unlike black swans, which are fundamentally unforeseeable (**?**), grey swans are physically plausible failures that sit in the long tail of operating experience. They present a data paradox for prognostics: *the events most critical to predict are precisely the ones with the least training data*. A bearing that runs for 10,000 hours before failing provides abundant healthy-state data but perhaps a single failure trajectory.

Modern predictive maintenance promises to catch these failures before they cascade into catastrophic downtime. Deep learning approaches for remaining useful life (RUL) prediction have shown strong results on benchmark datasets (**??**), with architectures ranging from CNN-LSTMs to Transformer-based models achieving impressive in-domain accuracy. However, these methods share two critical limitations. First, they require extensive labeled failure data - run-to-failure trajectories where degradation is observed from inception to failure - which is expensive, dangerous, and scarce by definition. Second, they rely on handcrafted spectral features (spectral centroid, kurtosis, envelope analysis) designed by domain experts, limiting applicability to well-studied component types.

Meanwhile, time series foundation models have emerged as a paradigm for general-purpose temporal representation learning. Models such as TimesFM (**?**), Chronos (**?**), and MOMENT (**?**) demonstrate strong zero-shot forecasting by pretraining on massive time series corpora. Yet these models solve a different problem: they forecast the *next value* in a sequence. For mechanical prognostics, predicting

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the next vibration sample is neither necessary nor sufficient - a perfect one-step-ahead predictor of raw waveform amplitude tells us nothing about remaining useful life. What matters is detecting the *spectral signature changes* that track degradation: the shift in spectral centroid as surface roughness increases, the redistribution of vibrational energy across frequency bands as fatigue cracks propagate. This distinction - *representing what matters for degradation* rather than reconstructing everything - motivates a targeted self-supervised approach designed for the structure of mechanical failure data.

We propose a self-supervised framework that addresses this data paradox by pretraining on abundant unlabeled operational data and requiring only a small number of failure trajectories for fine-tuning. The framework has three components:

1. **JEPA pretraining** learns waveform texture and periodicity from unlabeled vibration data across multiple bearing datasets, using masked patch prediction in a joint embedding predictive architecture (**?**). The encoder learns to represent vibration signals without any degradation labels.
2. **Temporal contrastive learning** exploits the natural ordering of run-to-failure episodes to learn what *changes* over a component's life. By training the encoder to distinguish early-life from late-life snapshots, it captures the spectral centroid shift that is the dominant degradation indicator for bearings.
3. **Hybrid fusion** combines self-supervised embeddings with domain-informed spectral features. Rather than treating self-supervised and handcrafted approaches as competing, we show they capture complementary information - JEPA learns fine-grained waveform features while spectral statistics capture the monotonic degradation trend.

A key finding is that different self-supervised objectives learn fundamentally different representations. JEPA's masked prediction objective captures waveform texture (PC1 correlation with spectral centroid: 0.071), while temporal contrastive learning captures spectral centroid shift (PC1 correlation: 0.856). This mechanistic understanding explains a practical trade-off: JEPA excels within-dataset where fine-grained waveform features transfer, while contrastive learning dominates cross-dataset transfer where only invariant degradation dynamics generalize.

Contributions.

1. A self-supervised framework for mechanical prognostics that learns degradation-aware representations from unlabeled vibration data across 8 bearing datasets (33,939 pretraining windows). JEPA pretraining alone matches expert-designed handcrafted features ($p = 0.40$); their fusion achieves RMSE 0.055 ± 0.004 - a 21.4% improvement over the best handcrafted-only method (**??**).
2. Cross-dataset RUL transfer via temporal contrastive learning without target-domain labels, achieving $+38\%$ over time-only baselines ($p < 0.001$) on FEMTO→XJTU-SY - the first SSL-based cross-dataset result for bearing RUL prognostics (**??**).
3. A mechanistic analysis showing that JEPA captures waveform texture while contrastive learning captures spectral centroid shift, explaining when and why each objective excels (**??**).
4. Generalization to multivariate sensor streams (Appendix **??**): a trajectory variant of JEPA applied to NASA C-MAPSS turbofan data achieves RMSE $= 14.23 \pm 0.39$ on FD001, within 1.7% of the best published SSL result (AE-LSTM: 13.99) while substantially outperforming a supervised LSTM at all five label budgets. Critically, we verify the result reflects genuine degradation tracking, not population-mean regression: the frozen encoder recovers the simulator's health index with $R^2 = 0.926$ without labels, and within-engine degradation correlation is $\rho = 0.830 \pm 0.023$ (5 seeds), with a $+41.5$ RMSE penalty when context vectors are shuffled across engines.

2 Related Work

2.1 Self-Supervised Learning for Time Series

Self-supervised pretraining for time series has followed two paradigms: contrastive and generative. Both have matured rapidly, yet neither has been adapted for mechanical prognostics.

Contrastive methods. **?** learn representations via hierarchical contrastive learning across temporal and instance dimensions. **?** exploit the stationarity of temporal neighborhoods, defining positive pairs within stationary windows - an assumption violated by bearing degradation, which is fundamentally non-stationary. **?** combine temporal and contextual contrasting, while **?** align time-domain and frequency-domain representations, demonstrating cross-domain transfer on medical and mechanical datasets. Of these, TF-C is closest to our setting, but evaluates on fault *classification*, not RUL regression.

Masked reconstruction. **?** adapt the Vision Transformer to time series via channel-independent patching and masked pretraining, achieving state-of-the-art long-term forecasting. **?** reconstruct masked time points from multiple masked versions to recover manifold structure, and **?** combine masked autoencoders with decoupled representation learning. These methods optimize for reconstruction fidelity, which biases representations toward low-level waveform statistics rather than degradation-relevant spectral features.

Foundation models. **?**, **?**, **?**, and **?** pretrain large models on diverse time series corpora for zero-shot or multi-task forecasting. These foundation models predict the *next value* - a fundamentally different objective from encoding degradation state. Our work targets this gap: rather than general-purpose time series modeling, we learn representations aligned with mechanical degradation indicators such as spectral centroid shift.

2.2 Remaining Useful Life Prediction

RUL prediction has progressed from physics-based models to deep learning. Early work applied CNNs (**?**) and LSTMs (**?**) to sensor data; modern architectures combine these with attention mechanisms. On C-MAPSS (**?**), the current state of the art is STAR (**?**), a two-stage attention Transformer with hierarchical patch merging (FD001 RMSE: 10.61). Architectures fusing Transformer, TCN, and self-attention branches such as TTS-Net (**?**) achieve FD001 RMSE: 11.02.

For bearing prognostics, FEMTO/PRONOSTIA (**?**) and XJTU-SY (**?**) are the primary benchmarks. Multi-scale separable convolution Transformers achieve RMSE 0.124 on FEMTO and 0.160 on XJTU-SY (**?**). **?** exploit time-invariant Neural ODE dynamics for cross-domain bearing transfer. However, nearly all methods share two limitations: (1) they require full temporal trajectory context (ordered sequences of windows), and (2) they are fully supervised or use domain adaptation with labeled target data. **?** present DCSSL, the only published SSL method for bearing RUL on FEMTO, using dual-dimensional contrastive pretraining with trajectory context. Our work differs in three respects: we use JEPA (not contrastive) pretraining, operate on single windows without trajectory context, and demonstrate cross-dataset transfer without explicit domain adaptation.

2.3 Joint Embedding Predictive Architectures

Joint Embedding Predictive Architectures (JEPA) learn by predicting representations of masked inputs from visible context in a shared latent space (**?**). **?** demonstrate with I-JEPA that predicting in representation space avoids the low-level reconstruction bias of masked autoencoders. **?** extend to video with 90% spatiotemporal masking, learning motion and dynamics representations. **?** apply JEPA to fMRI brain dynamics with domain-specific positioning and masking (NeurIPS 2024 Spotlight), the closest analogue to our work in adapting JEPA to a non-vision domain with physics-informed structure.

For time series, **?** adapt JEPA with temporal masking for classification benchmarks, while **?** introduce codebook regularization for multivariate series, preventing representation collapse through discrete bottlenecks. **?** replace heuristic collapse prevention (EMA, stop-gradient) with the theoretically grounded SIGReg regularizer, proving that the optimal embedding distribution is isotropic Gaussian. None of these time series JEPA variants evaluate on prognostics or RUL tasks.

Two recent works apply self-supervised pretraining to vibration data. **?** use GPT-style generative pretraining on rotating machinery signals (68.5M parameters), achieving 92% one-shot fault classification but predicting raw signal tokens rather than latent representations. **?** apply masked autoencoders to 5M vibration samples from seismic sensors, showing that large-scale vibration pretraining transfers to downstream sensing tasks. However, *neither applies JEPA to mechanical systems, and neither*

evaluates on RUL prediction. Our work is the first to evaluate JEPA for mechanical RUL prediction, diagnose why standard JEPA representations have low correlation with degradation indicators (spectral centroid shift), and demonstrate that hybrid architectures overcome this limitation.

2.4 Learning with Limited Failure Data

The scarcity of failure data is a central challenge in prognostics (**?**). Grey swan events - rare but physically plausible failures - are well-studied in weather prediction, where AI models consistently fail on out-of-distribution extremes (**?**), and in supply chain risk, where formal frameworks distinguish grey swans (structurally predictable) from black swans (epistemically unknowable) (**?**). **?** generate synthetic grey swans via gradient-based optimization through differentiable physics models. In mechanical prognostics, **?** demonstrate competitive RUL prediction with zero real failure data using physics-informed augmentation, while **?** combine GANs with inverse PINNs to generate physically valid rare fault signals. **?** achieve $> 94\%$ fault classification with only 5 labeled real samples by pretraining on physics-simulated data. Reviews of physics-informed RUL methods (**?**) consistently find that physics constraints provide the largest benefit in low-data regimes - precisely the grey swan scenario. Our self-supervised pretraining approach is complementary: by learning general vibration representations from abundant unlabeled data (including healthy operation), we reduce the labeled data requirement to 18 failure trajectories, without requiring a physics simulator or handcrafted augmentation strategies.

3 Method

We present a self-supervised framework for mechanical RUL prediction comprising three stages: JEPA pretraining on unlabeled vibration data (**??**), temporal contrastive learning using run-to-failure episode structure (**??**), and hybrid fusion with domain-informed spectral features (**??**).

3.1 Problem Formulation

Consider a set of run-to-failure episodes $\{(\mathbf{x}_i^{(1)}, \ldots, \mathbf{x}_i^{(T_i)})\}_{i=1}^N$, where $\mathbf{x}_i^{(t)} \in \mathbb{R}^L$ is a vibration window of $L = 1024$ samples at time step t, and T_i is the total lifetime of episode i. We define the normalized remaining useful life as:

$$\text{RUL}(t) = 1 - \frac{t}{T_i}, \tag{1}$$

so RUL $= 1$ at the start of operation and RUL $= 0$ at failure. This normalization is essential because bearing lifetimes vary by up to $5\times$ across episodes and datasets.

The grey swan challenge is that failure episodes are scarce (we have $N = 23$ across two datasets), while unlabeled healthy operation data is abundant. Our framework addresses this asymmetry through self-supervised pretraining.

3.2 JEPA Pretraining

Our JEPA encoder follows the architecture of I-JEPA (**?**) adapted for 1D vibration signals.

Patch embedding. Each window $\mathbf{x} \in \mathbb{R}^L$ is divided into $P = 16$ non-overlapping patches of length $L/P = 64$ samples. Each patch is projected to a d-dimensional embedding via a linear layer, yielding patch tokens $\{p_1, \ldots, p_P\} \in \mathbb{R}^d$ with $d = 256$.

Encoder. The context encoder f_θ is a Vision Transformer (ViT) with 4 layers, 4 attention heads, and hidden dimension $d = 256$ (3.2M parameters). Sinusoidal positional encodings are added to all patch tokens.

Masking. We randomly mask 62.5% of patches (10 out of 16), providing only 6 context patches to the encoder. This high mask ratio is critical: at lower ratios ($\leq 50\%$), the predictor can exploit a context-average shortcut, leading to representation collapse (**??**).

Target encoder. The target encoder $\bar{f}_\xi$ is an exponential moving average (EMA) copy of f_θ with momentum $\tau = 0.996$:

$$\xi \leftarrow \tau\xi + (1-\tau)\theta. \tag{2}$$

Predictor. The predictor g_ϕ is a 4-layer Transformer ($d_{\text{pred}} = 128$, 0.9M parameters) that takes context encoder outputs and sinusoidal positional queries for masked positions, predicting the target encoder's representations.

Training objective. We minimize the L1 distance between predicted and target representations, with variance regularization to prevent collapse:

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|M|}\sum_{j\in M} \|g_\phi(f_\theta(\mathbf{x}_{\bar{M}}))_j - \text{sg}[\bar{f}_\xi(\mathbf{x})_j]\|_1 + \lambda \cdot \mathcal{L}_{\text{var}}, \tag{3}$$

where M is the set of masked indices, $\bar{M}$ the context indices, and sg$[\cdot]$ is stop-gradient. The variance regularization $\mathcal{L}_{\text{var}} = \max(0, 1 - \sqrt{\text{Var}(\mathbf{z}) + \epsilon})$ with $\lambda = 0.1$ prevents representation collapse by penalizing the encoder when embedding variance drops below a threshold - without this term, the encoder can minimize L1 loss by mapping all inputs to the same constant representation.

Pretraining data. We pretrain on 33,939 vibration windows from 8 bearing datasets (CWRU (**?**), MFPT, IMS (**?**), FEMTO (**?**), XJTU-SY (**?**), Paderborn, Ottawa, MAFAULDA), spanning diverse operating conditions, fault types, and sampling rates (all resampled to 12,800 Hz).

3.3 Temporal Contrastive Learning

While JEPA learns waveform texture through reconstruction, it does not capture the spectral changes that indicate degradation (see **??**). We address this with temporal contrastive learning that exploits the natural ordering of run-to-failure episodes.

Triplet objective. For each anchor window $\mathbf{x}_i^{(t)}$, we sample a positive from a nearby time step $\mathbf{x}_i^{(t')}$ with $|t - t'| < \delta_{\text{pos}}$ and a negative from a distant time step $\mathbf{x}_i^{(t'')}$ with $|t - t''| > \delta_{\text{neg}}$:

$$\mathcal{L}_{\text{contrastive}} = \max(0, \|f_\theta(\mathbf{x}^{(t)}) - f_\theta(\mathbf{x}^{(t')})\|_2 - \|f_\theta(\mathbf{x}^{(t)}) - f_\theta(\mathbf{x}^{(t'')})\|_2 + \alpha), \tag{4}$$

where $\alpha = 1.0$ is the margin. Intuitively, the triplet loss pulls together representations of temporally nearby windows (within δ_{pos} steps) and pushes apart representations of temporally distant windows (beyond δ_{neg} steps). This forces the encoder to learn what *changes* over the bearing's life - primarily the spectral centroid shift as surface roughness increases with degradation.

Training. We train on 18 run-to-failure episodes (75% training split) for 100 epochs using the temporal labels (not RUL targets). The encoder converges to positive-pair similarity 0.89 and negative-pair similarity 0.47, indicating strong temporal structure in the learned representation.

3.4 Hybrid Architecture

Our key finding is that JEPA and handcrafted features capture complementary information (**??**). The hybrid architecture combines both:

Feature extraction. For each vibration window, we extract:

- **JEPA embeddings**: the mean-pooled output of the frozen pretrained encoder, $\mathbf{h}_{\text{JEPA}} \in \mathbb{R}^{256}$.
- **Handcrafted features**: 8 time-domain statistics (RMS, peak, crest factor, kurtosis, skewness, shape factor, impulse factor, clearance factor), 7 frequency-domain features (spectral centroid, spread, entropy, and 4 band energies), and 3 envelope features (Hilbert-transform RMS, kurtosis, peak), yielding $\mathbf{h}_{\text{HC}} \in \mathbb{R}^{18}$.

Fusion. For each time step in the episode, the concatenated input $[\mathbf{h}_{\text{JEPA}}; \mathbf{h}_{\text{HC}}; \Delta t] \in \mathbb{R}^{275}$ is projected to $d_{\text{model}} = 128$ and processed by a Transformer encoder (4 layers, 4 heads) with sinusoidal positional encoding over the temporal sequence. A linear head maps the final representation (concatenated with normalized elapsed time) to a scalar RUL prediction.

Table 1: Dataset summary. Episodes are complete run-to-failure sequences from initial operation to bearing failure.

Dataset	Episodes	Sampling Rate	Channels	Lifetime Range	Operating Conditions
FEMTO (PRONOSTIA)	16	25.6 kHz	1 (vibration)	1–7 h	3 load/speed settings
XJTU-SY	7	25.6 kHz	2 (H+V vib.)	0.5–2.6 h	3 load/speed settings
Combined	**23**	-	-	**CV=0.635**	**6 conditions**

Table 2: In-domain RUL prediction results (mixed FEMTO+XJTU-SY, 5-seed average). Improvement is relative reduction in RMSE vs. elapsed-time-only baseline (RMSE=0.224).

Category	Method	RMSE	±std	vs. Time-Only
Naive	Constant mean	0.290	-	-29.4%
	Elapsed time only	0.224	-	0%
Signal proc.	Envelope RMS + LSTM	0.287	0.001	-28.1%
End-to-end	CNN-LSTM	0.195	0.005	$+13.0\%$
	CNN-GRU-MHA	0.185	0.005	$+17.4\%$
Handcrafted	HC + LSTM	0.177	0.016	$+21.2\%$
	HC + MLP	0.085	0.004	$+62.0\%$
	Transformer + HC	0.070	0.006	$+68.9\%$
Self-supervised	Random JEPA + LSTM	0.221	0.008	$+1.5\%$
	JEPA + LSTM	0.189	0.015	$+15.8\%$
Hybrid	**JEPA + HC Transformer**	**0.055**	**0.004**	$\mathbf{+75.5\%}$

Training. The fusion model is trained end-to-end with MSE loss on 75% of episodes, with JEPA encoder weights frozen. We report mean ± std over 5 seeds on held-out episodes.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on two bearing run-to-failure datasets with complementary characteristics (**??**):

Evaluation protocol. All methods are evaluated with 75%/25% episode-based train/test splits, fixed across methods. We report RMSE on normalized RUL ($\in [0, 1]$) averaged over 5 seeds (10 seeds for cross-dataset experiments) with standard deviations. Statistical significance is assessed via paired t-tests.

Baselines. We compare 11 methods spanning naive baselines, end-to-end deep learning, handcrafted features, and self-supervised approaches:

- **Naive**: Constant mean predictor, elapsed-time-only baseline.
- **Signal processing**: Envelope RMS + LSTM.
- **End-to-end**: CNN-LSTM, CNN-GRU-MHA.
- **Handcrafted**: HC+LSTM, HC+MLP, Transformer+HC.
- **Self-supervised**: Random JEPA+LSTM (untrained encoder), JEPA+LSTM (pretrained).
- **Hybrid**: JEPA+HC Transformer (our best).

4.2 In-Domain RUL Prediction

???? present the complete comparison. Several findings emerge:

figures/v8_rul_comparison.pdf

Figure 1: In-domain RUL prediction comparison across all methods. The hybrid architecture (JEPA + HC Transformer) achieves RMSE 0.055 ± 0.004, a 75.5% improvement over the elapsed-time baseline. Error bars show ± 1 std over 5 seeds.

JEPA pretraining is effective. JEPA+LSTM (RMSE 0.189) significantly outperforms both the elapsed-time baseline ($+15.8\%$, $p = 0.010$) and random encoder initialization ($p = 0.010$). The pretrained encoder learns meaningful vibration representations without any RUL labels.

JEPA matches handcrafted features. The difference between JEPA+LSTM (0.189) and HC+LSTM (0.177) is not statistically significant ($p = 0.40$), demonstrating that self-supervised pretraining can match decades of domain expertise in feature engineering - without requiring that expertise.

The hybrid wins decisively. Combining JEPA embeddings with handcrafted features in a Transformer architecture achieves the best result: RMSE 0.055 ± 0.004, a 75.5% improvement over elapsed-time and 21.4% over the best handcrafted-only method (Transformer+HC at 0.070). This confirms that JEPA captures complementary information to spectral statistics.

Table 3: Cross-dataset transfer results (10-seed average). Elapsed time baseline degrades sharply cross-dataset due to lifetime distribution mismatch.

Configuration	Elapsed Time	JEPA+LSTM	Contrastive+LSTM	Contrastive vs. JEPA
FEMTO→FEMTO (within)	0.027	0.113 ± 0.011	0.142 ± 0.012	-20.4%
XJTU→XJTU (within)	0.159	0.195	0.214	-9.7%
FEMTO→XJTU (cross)	**0.367**	0.280 ± 0.007	$\mathbf{0.227 \pm 0.015}$	$+18.8\%$ ($p < 0.001$)
XJTU→FEMTO (cross)	**0.336**	0.403	$\mathbf{0.309 \pm 0.007}$	$+23.2\%$

Table 4: Encoder representation quality. PC1 correlation measures how well the first principal component of the encoder's output aligns with RUL and the dominant degradation indicator (spectral centroid).

Metric	JEPA	Contrastive	Handcrafted
PC1 correlation w/ RUL	0.186	**0.648**	0.585
PC1 correlation w/ spectral centroid	0.071	**0.856**	1.000
Max per-dim correlation w/ RUL	0.144	**0.648**	0.585

4.3 Cross-Dataset Transfer

Cross-dataset transfer - training on one bearing test rig and evaluating on another - is the critical test of learned representations. Different test rigs have different operating conditions, bearing types, and lifetime distributions (coefficient of variation = 0.635).

???? show the cross-dataset results.

Contrastive dominates cross-dataset. Temporal contrastive learning significantly outperforms JEPA for cross-dataset transfer ($+18.8\%$ on FEMTO→XJTU, $p < 0.001$). This is because the contrastive objective forces the encoder to learn what *changes* over a bearing's life - the spectral centroid shift - which is the invariant degradation signature across datasets.

JEPA wins within-dataset. Conversely, JEPA achieves lower RMSE within datasets, especially on FEMTO (0.113 vs 0.142). JEPA's patch prediction objective preserves fine-grained waveform details that are useful when train and test distributions match.

Both beat elapsed time. In the cross-dataset setting, where elapsed time baselines degrade sharply (RMSE 0.336–0.367), both learned representations provide substantial improvement ($+24\%$ for JEPA, $+38\%$ for contrastive vs time-only).

5 Analysis and Discussion

5.1 What Do Self-Supervised Encoders Learn?

A central question is *why* different self-supervised objectives learn different representations, and how this explains their downstream performance (**??**).

JEPA learns waveform texture. The JEPA objective - predicting masked patches from context - optimizes for local waveform reconstruction. The encoder must capture periodicity, amplitude modulation, and patch-level temporal structure. However, these features have low correlation with the spectral centroid (0.071), which is the primary degradation indicator for bearings. This explains why JEPA alone underperforms handcrafted spectral features for RUL prediction. **??** shows JEPA latent trajectories for six FEMTO bearing episodes: while individual bearing episodes show directional drift (arrows), the scatter in PC space (PC1 explains only 9.7% of variance) reflects the waveform-texture nature of the representations rather than a clean degradation axis.

Contrastive learning captures degradation dynamics. The temporal contrastive objective - distinguishing early from late snapshots - forces the encoder to learn what *changes* over a bearing's

figures/v8_cross_dataset.pdf

Figure 2: Cross-dataset transfer results. Contrastive pretraining consistently outperforms JEPA for cross-dataset generalization ($+18.8\%$ on FEMTO→XJTU-SY, $p < 0.001$), while JEPA excels within-dataset. Both methods substantially outperform the elapsed-time baseline in the cross-dataset setting.

life. The dominant change is spectral centroid shift (correlation 0.856 with PC1), which occurs as surface roughness increases and distributes vibrational energy across frequencies (**??**). This alignment with the degradation indicator explains the strong cross-dataset transfer: spectral centroid shift is a physics-invariant degradation signature.

Hybrid complementarity. JEPA and contrastive representations are nearly orthogonal in their information content. JEPA captures fine-grained waveform features that are dataset-specific but informative within domain. Contrastive captures the slow, monotonic degradation trajectory that transfers across domains. The hybrid architecture leverages both, achieving the best overall performance.

5.2 When Does Transfer Succeed?

The divergent within-dataset vs. cross-dataset performance reveals a fundamental trade-off:

- **Within-dataset**: JEPA is preferred because fine-grained waveform features are discriminative when train/test distributions match. JEPA achieves RMSE 0.113 on FEMTO→FEMTO vs. 0.142 for contrastive.

figures/v8_encoder_analysis.pdf

Figure 3: PC1 of JEPA vs. contrastive encoder representations, plotted against spectral centroid and RUL%. JEPA PC1 is uncorrelated with spectral centroid ($r = 0.071$) and weakly correlated with RUL ($r = 0.186$). Contrastive PC1 strongly tracks spectral centroid ($r = -0.856$) and RUL ($r = -0.648$), confirming that the temporal contrastive objective learns degradation-relevant features while JEPA learns waveform texture.

- **Cross-dataset**: Contrastive is preferred because only dataset-invariant features (spectral centroid shift) transfer. Contrastive achieves RMSE 0.227 on FEMTO→XJTU vs. 0.280 for JEPA.

The coefficient of variation in episode lifetimes (CV = 0.635) explains why elapsed-time baselines fail cross-dataset: FEMTO bearings live 1–7 hours while XJTU-SY bearings live 0.5–2.6 hours, making calendar time a poor proxy for degradation state.

5.3 Limitations

1. **Small dataset.** With only 23 bearing episodes, our results have non-trivial variance (e.g., JEPA+LSTM std = 0.015). Each run-to-failure episode requires physically destroying a bearing, fundamentally limiting dataset scale.
2. **Single channel.** Current experiments use only one vibration channel. Extending to multivariate inputs (e.g., multi-axis accelerometers, motor current) is a natural next step.

figures/v8_latent_trajectories.pdf

Figure 4: JEPA V8 latent trajectories in PC space for six FEMTO bearing episodes. Color indicates RUL% (green=healthy, red=failed). Arrows show mean drift direction from healthy to failed state. The scatter (PC1 only 9.7% variance) confirms JEPA captures high-dimensional waveform structure rather than a single degradation axis - complementary to the contrastive encoder's clean spectral centroid signal.

3. **Temporal contrastive requires episode structure.** The contrastive objective assumes access to run-to-failure episodes with temporal ordering.
4. **JEPA pretraining instability.** Training loss oscillates after epoch 2 on heterogeneous multi-source data; the best checkpoint is always early (**??**, Appendix).
5. **Two transfer pairs.** Cross-dataset evaluation is limited to FEMTO↔XJTU-SY. A trajectory variant of JEPA applied to C-MAPSS turbofan data (Appendix **??**) confirms the framework generalizes to 14-channel multivariate sensor streams, but cross-domain bearing↔turbofan transfer is not evaluated.
6. **Gap to supervised SOTA.** On C-MAPSS FD001, JEPA E2E achieves RMSE 14.23 ± 0.39 versus STAR (supervised, RMSE 12.19 replication). Closing this $\sim$2 RMSE gap is ongoing work. Health index recovery ($R^2 = 0.926$) and within-engine tracking ($\rho = 0.830$) confirm the representations are genuinely learning degradation structure.

6 Conclusion

We introduced a self-supervised framework for mechanical prognostics in the grey swan regime - where the component failures most critical to predict are precisely those with the least training data. By combining JEPA pretraining (which captures waveform texture) with temporal contrastive learning (which captures degradation dynamics), and fusing with domain-informed spectral features, we achieve RMSE 0.055 ± 0.004, a 75.5% improvement over time-only baselines.

Our mechanistic analysis reveals that different self-supervised objectives learn fundamentally different representations, explaining their complementary strengths: JEPA excels within-dataset while contrastive learning enables cross-dataset transfer ($+38\%$ vs. time-only, $p < 0.001$). This understanding motivates the hybrid architecture and provides guidance for practitioners choosing between pretraining strategies.

Several directions remain open. Extending the framework to multivariate sensor inputs with physics-aware masking could exploit cross-modal dependencies. A trajectory variant of JEPA applied to C-MAPSS turbofan data (**??**) shows the framework generalizes to 14-channel sensor streams, substantially outperforming a supervised LSTM at all label budgets and matching the only prior SSL baseline (AE-LSTM) within 1.7%; further validation on additional mechanical domains would strengthen these claims. Finally, synthetic data augmentation using physics-informed degradation models could address the fundamental scarcity of rare failure trajectories that motivates this work.

A Dataset Details

The FEMTO bearing dataset (**?**) was collected at the FEMTO-ST institute using an accelerated degradation test platform. Bearings are subjected to radial loads while rotating at controlled speeds, with vibration measured by a piezoelectric accelerometer at 25.6 kHz. The dataset contains 17 run-to-failure experiments across 3 operating conditions (we use 16 with complete recordings).

The XJTU-SY bearing dataset (**?**) contains 15 rolling element bearings tested under 3 operating conditions. Each bearing is monitored with two accelerometers (horizontal and vertical) at 25.6 kHz. We use 7 bearings with clear degradation trajectories.

JEPA pretraining uses 33,939 vibration windows from 8 sources, all resampled to 12,800 Hz:

Table 5: Pretraining data sources.

Source	Windows	Sampling Rate	Fault Types
CWRU	2,400	12 kHz	Healthy, IR, OR, Ball
MFPT	1,200	25.6 kHz	Healthy, IR, OR
IMS	3,000	20 kHz	Run-to-failure
FEMTO	8,000	25.6 kHz	Run-to-failure
XJTU-SY	4,500	25.6 kHz	Run-to-failure
Paderborn	6,000	64 kHz	Healthy, IR, OR
Ottawa	4,839	12 kHz	Healthy, IR, OR, Ball
MAFAULDA	4,000	50 kHz	Multiple

B Hyperparameters

Table 6: Complete hyperparameter settings.

Component	Hyperparameter	Value
JEPA Pretraining	Encoder dim / depth / heads	256 / 4 / 4
	Predictor dim / depth	128 / 4
	Mask ratio	62.5%
	EMA momentum	0.996
	Learning rate	1×10^{-4}
	Batch size / Epochs	256 / 100 (best ep. 2)
	Loss	L1 + var. reg ($\lambda = 0.1$)
Contrastive	Learning rate / Margin α	1×10^{-4} / 1.0
	Epochs / Episodes	100 / 18
RUL Heads	LSTM: hidden / layers / dropout	128 / 2 / 0.2
	Transformer: layers / heads / dim	4 / 4 / 128
	Learning rate	1×10^{-3}

C Additional Results

Table 7: Linear vs. piecewise-linear RUL labels.

Method	Linear RMSE	Piecewise RMSE	Δ
JEPA + LSTM	0.189	0.310	+0.121
HC + MLP	0.085	0.114	+0.029
HC + LSTM	0.177	0.214	+0.037
Transformer + HC	0.070	0.114	+0.044

figures/v8_pretrain_history.pdf

Figure 5: JEPA V8 pretraining loss curve. The model achieves its best validation loss at epoch 2 (0.0168), after which training and validation losses diverge. This instability on heterogeneous multi-source data is a known limitation: the best checkpoint must be selected early, preventing full benefit from extended pretraining. We address this with early stopping based on probe RMSE at epoch 50.

Table 8: JEPA pretraining configurations explored.

Config	LR	EMA	Mask	Best Val Loss	Verdict
Default	10^{-4}	0.996	62.5%	0.0168 (ep. 2)	**Best**
Lower LR	3×10^{-5}	0.996	62.5%	0.0172	Similar
Fixed LR	10^{-5}	0.996	62.5%	0.0189	Worse
High EMA	10^{-4}	0.999	62.5%	0.0178	Worse
2ch FFT	10^{-4}	0.996	62.5%	0.0149	Better loss, worse RUL

D Extension to Turbofan RUL Prediction (C-MAPSS)

To assess whether self-supervised JEPA representations generalize beyond single-channel vibration to multivariate sensor streams, we apply a trajectory variant of JEPA to the NASA C-MAPSS turbofan dataset (**?**).

Trajectory JEPA for multivariate sensors. Instead of masking spatial patches in a 1D vibration window, we use a *trajectory* formulation: given past sensor observations $\mathbf{x}_{1:t}$, predict the latent representation of future observations $\mathbf{x}_{t+1:t+k}$ using a causal context encoder and an EMA target encoder. This captures temporal degradation dynamics across 14 sensor channels simultaneously, without requiring failure labels.

Dataset. C-MAPSS FD001 contains 100 training engines and 100 test engines, each operating under a single condition until failure. We select 14 informative sensors (removing near-constant channels), normalize using train-set statistics, and cap RUL at 125 cycles. We use an 85%/15%

train/val split by engine. All results are evaluated using the canonical test set (last-window-per-engine, RUL from `RUL_FD001.txt`).

Architecture. The context encoder is a 2-layer causal Transformer ($d_{\text{model}} = 256$, 4 heads, 1.26M parameters). Pretraining uses horizon-aware prediction: the encoder predicts latent representations $k \in [5, 30]$ steps ahead, with k sampled uniformly. An EMA target encoder (momentum $= 0.99$) provides stable prediction targets, preventing collapse. The model is pretrained for up to 200 epochs using probe-based early stopping (best probe RMSE at epoch 50).

Fine-tuning. A linear RUL probe is trained on frozen encoder representations (frozen mode) or jointly with the encoder (E2E mode). We evaluate at five label budgets (100%, 50%, 20%, 10%, 5% of training engines) using 5 seeds each.

Table 9: C-MAPSS FD001 RUL prediction. RMSE in cycles (RUL cap = 125). Mean $\pm$ std over 5 seeds. Lower is better. STAR paper result from **?**; STAR (replic.) is our 5-seed replication (FD001 config, same hyperparameters). AE-LSTM from **?**.

Type	Method	100%	50%	20%	10%	5%
Supervised	STAR (paper)	10.61	—	—	—	—
Supervised	STAR (replic.)	$12.19_{\pm 0.6}$	—	—	—	—
SSL (paper)	AE-LSTM	13.99	—	—	—	—
Supervised	LSTM (ours)	$17.36_{\pm 1.2}$	$18.30_{\pm 0.8}$	$18.55_{\pm 0.8}$	$31.22_{\pm 10.9}$	$33.08_{\pm 9.6}$
SSL (ours)	Traj JEPA Frozen	$17.81_{\pm 1.7}$	$18.71_{\pm 1.1}$	$19.83_{\pm 0.3}$	$19.93_{\pm 0.9}$	$21.53_{\pm 2.0}$
SSL (ours)	**Traj JEPA E2E**	$\mathbf{14.23}_{\pm 0.4}$	$\mathbf{14.93}_{\pm 0.4}$	$\mathbf{16.54}_{\pm 0.8}$	$\mathbf{18.66}_{\pm 0.8}$	$25.33_{\pm 5.1}$

Results.

Key findings.

1. **Competitive with prior SSL on C-MAPSS**: Traj JEPA E2E (14.23 ± 0.39 across 5 seeds) matches AE-LSTM SSL (13.99) within 1.7% - the only published SSL result on C-MAPSS FD001 - without using failure-time labels during pretraining. Unlike AE-LSTM, we evaluate at five label budgets, demonstrating consistent advantage over a supervised LSTM from 10% to 100% labels.
2. **Outperforms supervised LSTM at all label budgets**: At 10% labels (8 training engines), JEPA E2E achieves RMSE 18.66 vs LSTM 31.22 — a 40% reduction. Frozen JEPA (19.93) also substantially outperforms LSTM (31.22).
3. **Genuine within-engine tracking** (**??**): We rigorously verify that the RMSE improvement reflects real degradation tracking, not population-mean regression. Median within-engine Spearman $\rho = 0.830 \pm 0.023$ across 100 test engines (5 seeds; 63/100 engines have $\rho > 0.7$ on average). Shuffling h_{past} context vectors across engines at inference time degrades RMSE from 13.98 to 55.45 ± 1.62 ($+41.5$ cycles, 5 seeds), confirming that each embedding is engine-specific and not a fixed bias.
4. **Health index recovery without labels**: A linear Ridge probe trained on frozen h_{past} embeddings recovers the simulator's health index with validation $R^2 = 0.926$ (target: 0.7), robust across three health index parameterizations ($R^2 \in [0.917, 0.926]$). The embedding space is dominated by a single degradation direction: PC1 explains 47.6% of variance and correlates with the health index at $|\rho| = 0.797$, while PC2 and PC3 have $|\rho| \leq 0.16$.
5. **Stability at low labels**: JEPA frozen std = 0.34–2.0 across all budgets vs LSTM std = 9–11, a 5–30$\times$ variance reduction critical for industrial deployment.

These results confirm that the JEPA framework generalizes beyond vibration signals: the trajectory prediction objective learns degradation-aware representations from unlabeled multivariate sensor streams, with strong label efficiency and stability advantages over supervised baselines. Crucially, the RMSE improvements reflect genuine within-engine degradation tracking rather than population-mean effects, as verified by five independent diagnostic checks (**????**).

Evaluation protocol transparency. The standard C-MAPSS evaluation protocol tests each engine at a single cut point (the last observed cycle), systematically selecting the hardest case where the

figures/v12_label_efficiency.pdf

Figure 6: C-MAPSS FD001 label efficiency comparison. JEPA E2E (solid blue) outperforms the supervised LSTM at all five label budgets. JEPA Frozen (dashed) outperforms LSTM at very low budgets ($\leq$10% labels, 8 engines), demonstrating critical advantages for the grey swan regime where failure trajectories are scarce. LSTM variance explodes below 20% labels (large error bars); JEPA frozen std remains $\leq$2.0 cycles across all budgets (5–30$\times$ more stable). Dashed horizontal lines show the AE-LSTM SSL baseline (13.99) and STAR supervised replication (12.19), both at 100% labels only.

engine is maximally degraded and history is minimal. We additionally report a sliding-cut evaluation, measuring RMSE across all cut points from cycle 30 to the end for each test engine. The sliding-cut RMSE is 11.77 versus 13.98 for last-window (a 16% improvement), with per-engine mean RMSE of 9.87. This is expected: at earlier cut points, the encoder has more history and can track degradation more precisely. We report both metrics. The last-window metric is standard and enables comparison to published work; the sliding-cut metric more honestly measures how the model performs across the full operational range.

FD002 distribution shift diagnosis. FD002 (six operating conditions, 260 training engines) shows a large val/test gap: the frozen encoder achieves val probe RMSE $= 15.35$ on in-distribution engines but 26.07 ± 0.26 on the canonical test set (**??**a). Condition analysis reveals the root cause (**??**b–c): operating conditions 1, 2, and 5 are overrepresented at test time relative to training (ratios of $2.5\times$, $2.6\times$, and $3.1\times$ respectively), while condition 0 (50% of training) is only 27% of test windows. The per-condition KMeans normalization in V11 is calibrated on the training distribution and mis-scales test windows that fall in rare training conditions. This is not an SSL pretraining failure - the encoder learns FD002 degradation well (val RMSE $= 15.35$) - but a test-distribution mismatch inherent in the C-MAPSS benchmark design.

figures/v12_cmapss_main_results.pdf

Figure 7: C-MAPSS FD001 verification summary. **(a)** Baseline hierarchy: constant predictor (43.3 RMSE), hand-engineered ridge regressor on per-engine summaries (19.2 RMSE), JEPA frozen (17.8 RMSE), and JEPA E2E (14.2 RMSE, 5-seed mean). Each step represents genuine within-engine tracking rather than population-mean fitting. **(b)** Health index recovery validation R^2 across three H.I. parameterizations; all exceed 0.7, confirming the result is not sensitive to the specific H.I. definition. **(c)** Sliding-cut versus last-window evaluation: the model performs 15% better when evaluated across all cut points (11.8 vs 14.0 RMSE), as expected from a genuine degradation tracker.

Cross-fault transfer (FD001 → FD003). To test whether representations transfer across fault modes, we pretrain on FD001 (fan blade degradation) and evaluate with a frozen linear probe on FD003 (HPC degradation, same operating condition). This is a zero-shot encoder transfer: the probe is re-trained with FD003 labels, but the encoder weights are fixed from FD001 pretraining.

The cross-fault transfer incurs a consistent 7–9 RMSE penalty relative to in-domain (Table **??**). The penalty reflects that FD001 and FD003 have different degradation signatures: fan blade wear and HPC wear affect different sensors and follow different temporal patterns. Despite this, the cross-fault frozen probe at 10% labels (RMSE = 28.79) still outperforms a fully supervised LSTM with the same label budget in-domain (RMSE = 31.22). This demonstrates that SSL representations provide useful structure even across mismatched fault modes.

Cross-subset pretraining (FD002 → FD001). We also test whether pretraining on FD002 (260 engines, 6 operating conditions) improves FD001 fine-tuning via a richer pretraining corpus. FD002-

figures/v12_cmapss_tracking.pdf

Figure 8: Within-engine degradation tracking evidence. **(a)** Distribution of per-engine prediction standard deviation across 100 FD001 test engines; mean median $= 12.1 \pm 0.7$ cycles over 5 seeds (far above the constant-predictor threshold of 3). **(b)** Distribution of within-engine Spearman ρ; mean median $= 0.830 \pm 0.023$ over 5 seeds, with 63/100 engines achieving $\rho > 0.7$ on average. **(c)** Shuffle test: shuffling h_{past} context vectors across engines at inference time degrades RMSE from 14.0 to 55.4 ($+41$ cycles, 5 seeds), proving the embeddings are engine-specific rather than fixed biases.

pretrained frozen probes match or slightly outperform FD001 in-domain frozen probes at high label counts (100%: 17.50 vs 17.81; 50%: 17.43 vs 18.71). However, at 10% labels the cross-subset frozen probe is substantially *worse* (24.45 vs 19.93). E2E fine-tuning consistently underperforms in-domain pretraining at all budgets (e.g., 100%: 17.50 vs 14.23). These results indicate that the FD002 encoder learned operating-condition-specific representations that resist adaptation to FD001's single-condition setting. More training data from a different distribution does not substitute for matched pretraining data, especially at low label budgets.

FD003 in-domain generalization. To verify the approach generalizes across C-MAPSS subsets, we also apply Trajectory JEPA to FD003, which has the same single operating condition as FD001 but combines two fault modes (fan blade and HPC degradation). We pretrain from scratch on FD003 training engines, then fine-tune at 100%, 20%, and 10% label budgets. Traj JEPA E2E achieves RMSE = $15.37_{\pm 0.9}$ at 100% labels, outperforming a supervised LSTM baseline (17.36, evaluated on FD001 with same architecture) on the harder mixed-fault subset. At 10% labels, JEPA E2E ($21.54_{\pm 1.5}$)

figures/v12_fd002_diagnosis.pdf

Figure 9: FD002 distribution shift diagnosis. **(a)** Val vs. test RMSE gap: FD001 has a -2.1 gap (test easier), while FD002 has a +10.7 gap. The frozen encoder generalizes within the training distribution (val=15.35) but fails on the canonical test set. **(b)** Per-condition test RMSE; conditions 1, 2, 3, 5 (red) are overrepresented at test time and show higher RMSE. **(c)** Training vs. test condition distributions: conditions 1, 2, 5 are $2.5\times$–$3.1\times$ overrepresented in test last-window cycles vs. training.

substantially outperforms the supervised LSTM reference ($31.22_{\pm 10.9}$, FD001), confirming the label efficiency advantage generalizes beyond single-fault-mode data.

FD004: multi-condition multi-fault challenge. FD004 is the hardest C-MAPSS subset, combining 6 operating conditions with 2 fault modes across 249 training engines. Trajectory JEPA E2E achieves RMSE = $25.62_{\pm 0.3}$ at 100% labels (vs STAR supervised 15.87 (**?**)). While E2E consistently outperforms frozen probe across all label budgets (25.62 vs 29.35 at 100%; 29.03 vs 31.08 at 10%), the absolute gap to supervised SOTA is substantially larger than on FD001/FD003. We attribute this to the increased representational demand: the V2 encoder (d_model=256) must learn operating-condition-invariant degradation features from 6 distinct operating regimes. Notably, prediction variance is very low (std $\leq$ 0.61 across all budgets), indicating stable representations even on this challenging subset. Scaling encoder capacity for multi-condition settings is a clear direction for future work.

Table 10: Cross-fault transfer: FD001-pretrained encoder evaluated on FD003 (different fault mode). RMSE in cycles. Mean $\pm$ std over 5 seeds. Lower is better. STAR paper result from **?**; STAR (replic.) is our 5-seed replication.

Setting	100%	50%	20%	10%	5%
FD001 in-domain (frozen)	17.81	18.71	19.83	19.93	21.53
FD001→FD003 cross-fault (frozen)	$24.79_{\pm 0.6}$	$25.91_{\pm 0.4}$	$27.82_{\pm 1.5}$	$28.79_{\pm 0.4}$	$28.28_{\pm 0.2}$
Supervised LSTM (FD001 reference architecture)	17.36	18.30	18.55	31.22	33.08
STAR (paper, FD003)	10.71	—	—	—	—
STAR (replic., FD003)	$12.74_{\pm 0.1}$	—	—	—	—